SAE J1939 Primer: What J1939 is.

SAE J1939 is a family of standards from the Society of Automotive Engineers for communication over CAN in heavy-duty vehicles, diesel engines, generator sets and agricultural and construction machinery. It builds a complete application layer on top of the CAN data link so that equipment from different manufacturers can exchange engine and drivetrain data.

J1939 networks use 29-bit extended CAN identifiers. The most common physical layer runs at 250 kbit/s on a shielded twisted pair, while a newer variant uses 500 kbit/s. Most engine data is broadcast periodically, and a request mechanism lets a device ask for a message that is not sent cyclically.

SAE J1939 Primer: Identifier structure and addressing.

The 29-bit J1939 identifier is split into fields: a 3-bit priority, a reserved bit, a data page bit, an 8-bit PDU Format (PF), an 8-bit PDU Specific (PS) field and an 8-bit source address (SA). The reserved bit, data page bit, PF and PS together form the 18-bit Parameter Group Number (PGN), which identifies what the message contains.

If the PF value is below 240 the message is of type PDU1 and the PS field holds the destination address, so the message is sent to one node. If PF is 240 or higher the message is PDU2, a broadcast message, and the PS field acts as a group extension. Source addresses are 8 bits wide: 254 is the null address and 255 is the global address.

Control messages usually default to priority 3, while informational and most other messages default to priority 6. A lower priority number wins arbitration. Devices obtain their source address with the address claiming procedure, which uses PGN 60928.

SAE J1939 Primer: Common engine messages and signal scaling.

Each value in a message is a Suspect Parameter Number (SPN). To convert a raw value, multiply by the resolution and add the offset. Several widely used messages are listed here.

Electronic Engine Controller 1 (EEC1) is PGN 61444 (0xF004). It carries engine speed as SPN 190 in bytes 4 and 5, with a resolution of 0.125 rpm per bit.

Engine Temperature 1 (ET1) is PGN 65262 (0xFEEE). Engine coolant temperature is SPN 110 in byte 1. It has a resolution of 1 degree C per bit and an offset of -40 degrees C, so a raw value of 130 means 90 degrees C.

Fuel Economy (LFE) is PGN 65266 (0xFEF2). The engine fuel rate is SPN 183 in bytes 1 and 2 with a resolution of 0.05 litres per hour per bit. Cruise Control and Vehicle Speed (CCVS) is PGN 65265 (0xFEF1) and carries wheel-based vehicle speed as SPN 84 in bytes 2 and 3 with a resolution of 1/256 km/h per bit.

SAE J1939 Primer: Diagnostics and long messages.

J1939 diagnostics use diagnostic messages (DMs). DM1 is PGN 65226 (0xFECA) and reports the currently active diagnostic trouble codes (DTCs). By default an ECU broadcasts DM1 once per second while a fault is active, and also sends it immediately when the fault status changes. DM2 (PGN 65227) lists previously active DTCs, and DM11 (PGN 65235) is used to clear active DTCs.

Each DTC contains a 19-bit SPN that names the faulty signal, a 5-bit Failure Mode Identifier (FMI) that describes the type of fault, a 7-bit occurrence count and a conversion method bit. DM1 also carries the lamp status: malfunction indicator lamp, red stop lamp, amber warning lamp and protect lamp.

A CAN frame holds only 8 data bytes, so J1939 defines a transport protocol for longer messages. A sender announces the transfer with the connection management message TP.CM (PGN 60416) and then sends the data in TP.DT frames (PGN 60160). The transport protocol can move up to 1785 bytes in one message.